

# Zhicheng (Stark) Guo

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I develop machine learning methods that turn **complex learned representations into human-understandable concepts** for **scientific discovery**. My work is primarily in machine learning methodology, particularly deep neural networks and foundation models. These methods have proven directly useful for scientific discovery. In particular, my work with radiologists develops an interpretable lexicon for a new type of imaging biomarker for breast cancer. My work with ICU neurologists provides the first data-driven support for the well-known ictal-interictal-injury continuum (IIIC) hypothesis. In my work on wearable devices for heart monitoring, I discovered a shared manifold across different modalities of cardiac signals that improves downstream detection of atrial fibrillation. My work is published in the Journal of Machine Learning Research (JMLR), CVPR, NEJM AI, Nature Machine Intelligence, and the Harvard Data Science Review.

## EDUCATION **Duke University**

Durham, NC

Ph.D. in Electrical and Computer Engineering  
(Advisor: Cynthia Rudin)

Jan 2022 – May 2026

M.S. in Computer Science (Transferred to Ph.D. Program)

Aug 2020 – Dec 2021

## **Rensselaer Polytechnic Institute**

Troy, NY

B.Sc. in Computer Science

Jul 2016 – May 2020

## PUBLICATIONS \* indicates co-first or co-senior authorship (equal contribution)

[Google Scholar](#)

† indicates mentored student

### **Selected Publications**

- [1] “ProtoCDisco: Unsupervised Concept Discovery with Foundation Model Representations”  
**Zhicheng Guo**, Maximillian Machado, Jon Donnelly, Cynthia Rudin. *Preprint*, 2026.
- [2] “What is Different Between These Datasets? A Framework for Explaining Data Distribution Shifts.” Varun Babbar\*, **Zhicheng Guo\***, Cynthia Rudin.  
*Journal of Machine Learning Research (JMLR)*, 2025. 
- [3] “SiamAF: Improving atrial fibrillation detection using a shared latent space for ECG and PPG signals.” **Zhicheng Guo**, Cheng Ding, Duc H. Do, Amit Shah, Randall J Lee, Xiao Hu, Cynthia Rudin.  
*Harvard Data Science Review (HDSR)*, 2025. 
- [4] “Rashomon Sets for Prototypical-Part Networks: Editing Interpretable Models in Real-Time.”  
Jon Donnelly, **Zhicheng Guo**, Alina Jade Barnett, Hayden McTavish, Chaofan Chen, Cynthia Rudin.  
*Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 
- [5] “Improving Clinician Performance in Classification of EEG Patterns on the Ictal-Interictal-Injury Continuum using Interpretable Machine Learning.” Alina Jade Barnett\*, **Zhicheng Guo\***, Jin Jing\*, Wendong Ge, Brandon Westover, Cynthia Rudin.  
*New England Journal of Medicine AI (NEJM AI)*, 2024. 
- [6] “Learned Kernels for Sparse, Interpretable, and Efficient Medical Time Series Processing.”  
Sully F Chen†, **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin.  
*Nature Machine Intelligence*, 2024. 
- [7] “A Supervised Machine Learning Semantic Segmentation Approach for Detecting Artifacts in Plethysmography Signals from wearables.” **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin.  
*Physiological Measurement*, 2021. 

## Other Publications

- [8] “Data-hugging shields proprietary AI models from research that could disprove them.” Anish Karpurapu<sup>†</sup>, **Zhicheng Guo**, Xiao Hu, Cynthia Rudin. *npj Artificial Intelligence*, 2026. 
- [9] “Cosine Similarity is Almost all you Need (for Prototypical-Part Models).” Luke Moffett, Frank Willard, Maximillian Machado, Emmanuel Mokel, Jon Donnelly, **Zhicheng Guo**, Adam Costarino<sup>†</sup>, Julia Yang, Giyoung Kim, Alina Jade Barnett, Cynthia Rudin. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. 
- [10] “It’s LIT! LLMs with Interpretable Tools.” Ruixin Zhang<sup>†</sup>, Jon Donnelly, **Zhicheng Guo**, Ghazal Khalighinejad, Haiyang Huang, Alina Jade Barnett, Cynthia Rudin. *Multi-Turn Interactions in Large Language Models Workshop (MTI-LLM Workshop, NeurIPS)*, 2025. 
- [11] “Learning From Alarms: A Robust Learning Approach for Accurate Photoplethysmography-Based Atrial Fibrillation Detection Using Eight Million Samples Labeled with Imprecise Arrhythmia Alarms.” Cheng Ding, **Zhicheng Guo**, Cynthia Rudin, Ran Xiao, Amit Shah, Duc H Do, Randall J Lee, Gari Clifford, Fadi B Nahab, Xiao Hu. *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2024. 
- [12] “SiamQuality: a ConvNet-based Foundation Model for Photoplethysmography Signals.” Cheng Ding, **Zhicheng Guo**, Zhaoliang Chen, Randall J Lee, Cynthia Rudin, Xiao Hu. *Physiological Measurement*, 2024. 
- [13] “Integrated Single-cell Multiomic Analysis of HIV Latency Reversal Reveals Novel Regulators of Viral Reactivation.” Manickam Ashokkumar, Wenwen Mei, Jackson J Peterson, Yuriko Harigaya, David M Murdoch, David M Margolis, Caleb Kornfein, Alex Oesterling<sup>†</sup>, **Zhicheng Guo**, Cynthia D Rudin, Yuchao Jiang, Edward P Browne. *Genomics, Proteomics & Bioinformatics*, 2024. 

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|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| PROFESSIONAL<br>EXPERIENCE | <b>Duke University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jul 2026 – Present  |
|                            | Postdoctoral Researcher — Foundation Models & Interpretability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |
|                            | <ul style="list-style-type: none"><li>- Developing vision foundation-model and representation-learning methods for breast cancer risk prediction from large-scale mammography data.</li><li>- Building an interpretable lexicon of mammographic texture concepts, together with scalable evaluation pipelines for high-stakes medical imaging.</li><li>- Developing unsupervised concept-discovery methods from vision foundation-model representations and evaluating the discovered concepts through human studies.</li></ul>                                                                                         |                     |
|                            | <b>Texas Instruments – Kilby Labs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | May 2025 – Aug 2025 |
|                            | Machine Learning Researcher Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |
|                            | <ul style="list-style-type: none"><li>- Architected and deployed a self-evolving LLM-agentic optimization framework for hardware-constrained neural architecture search and mixed-precision quantization on edge hardware.</li><li>- Implemented multi-objective reward functions and on-the-fly hardware profiling to guide optimization decisions and improve real-time performance.</li><li>- Developed specialized small reasoning language models (SLMs) as replacements for generic LLM agents with GRPO and fine-tuning, enabling on-site and data-secure inference at a fraction of the compute cost.</li></ul> |                     |
|                            | <b>Duke University Health System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | May 2021 – Aug 2021 |
|                            | Quantitative Research Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |
|                            | <ul style="list-style-type: none"><li>- Investigated correlations between cellular populations/markers in CSF and PBMC with cognitive scores in HIV-associated neurocognitive disorder patients under different drug use conditions.</li></ul>                                                                                                                                                                                                                                                                                                                                                                          |                     |

- Developed machine learning models with flow cytometry data to predict drug use and impairment, enabling clinical researchers to assess drug effects on patients.

**PATENTS    Texas Instruments – Kilby Labs**

- [1] LLM-guided neural network architecture and quantization search for TINIE (Texas Instruments Neural Inference Engine) on microcontrollers. (pending)
- [2] Small language models for TINIE-optimized edge AI model development. (pending)

**MEDIA    OP-EDS**

- [1] Cynthia Rudin, **Zhicheng Guo**, Cheng Ding, and Xiao Hu. “How good are AI health technologies? We have no idea.” *STAT News*, Oct. 11, 2023. 

**Media Coverage**

- [2] “AI helps medical professionals read confusing brain EEGs” *Mindplex*, May 29, 2024. 
- [3] “New AI tool from Duke researchers reads EEGs, could improve treatment of seizures” *Yahoo News (The News & Observer)*, Feb. 20, 2025. 

**SERVICE    Reviewer**

|                                                   |                        |
|---------------------------------------------------|------------------------|
| IEEE Journal of Biomedical and Health Informatics | 2023, 2024, 2025, 2026 |
| Physiological Measurement                         | 2025                   |
| Epilepsia                                         | 2024                   |
| NeurIPS                                           | 2024, 2025, 2026       |

**TEACHING    Duke University**

|                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| Graduate Theory and Algorithms for Machine Learning 671D                   | Fall 2025 |
| - Guest lecture on neural networks and interpretable neural network design |           |
| TA, Graduate Theory and Algorithms for Machine Learning 671D               | Fall 2023 |
| TA, Graduate Theory and Algorithms for Machine Learning 671D               | Fall 2022 |
| TA, Graduate Theory and Algorithms for Machine Learning 671D               | Fall 2021 |

**Rensselaer Polytechnic Institute**

|                                |             |
|--------------------------------|-------------|
| SAPRIS Program Academic Mentor | Summer 2017 |
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**TALKS    INFORMS**

|                                                    |                             |
|----------------------------------------------------|-----------------------------|
| Invited talk on Interpretable Deep Learning Models | Atlanta, GA<br>October 2025 |
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**Cornell ORIE Young Researchers Workshop**

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| Invited poster | Ithaca, NY<br>October 2024 |
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**ADVISED    Will Li — Master’s student, Duke University**

**STUDENTS    Anish Karpurapu — MD-PhD student, Duke University**

**Liam Connor — Master’s student, Duke University**

**Pranay Jain — Former master’s student, now PhD student, University of Washington**

**Dennis Tang — Undergraduate student, incoming PhD student, Duke University**

**Alex Oesterling — Former undergraduate student, now PhD student, Harvard University**

**Adam Costarino — Former master’s student, now in industry**

**Bhrij Patel — Former undergraduate student, now PhD student, University of Maryland**

**Sully F Chen — Former MD-PhD student, now at OpenAI**

**Ruixin Zhang — Former master’s student, now at Google**

**Haoyu (Cindy) Wu — Former master’s student, now in industry**